

CALCULATION · UTILITY

Piping Calculators (ACEUTIL)

Small field calculators: fitting weight, pressure drop (steam/water/general), pipe thickness, insulation density.

Key Purpose

Why this program

1. Field calculators in one small suite

– Fitting weight · steam/water/general pressure drop · pipe thickness · insulation density — each a quick standalone tool

→ **Effect:** Routine piping checks are answered on the spot, without opening a spreadsheet or hand-deriving formulas

2. Pick pipe thickness & weight from spec tables

– Scans Excel spec files in the folder, auto-detects the size / SCH / thickness / weight columns even in messy sheets

→ **Effect:** No manual column hunting — any vendor or standard table drops in and is instantly usable

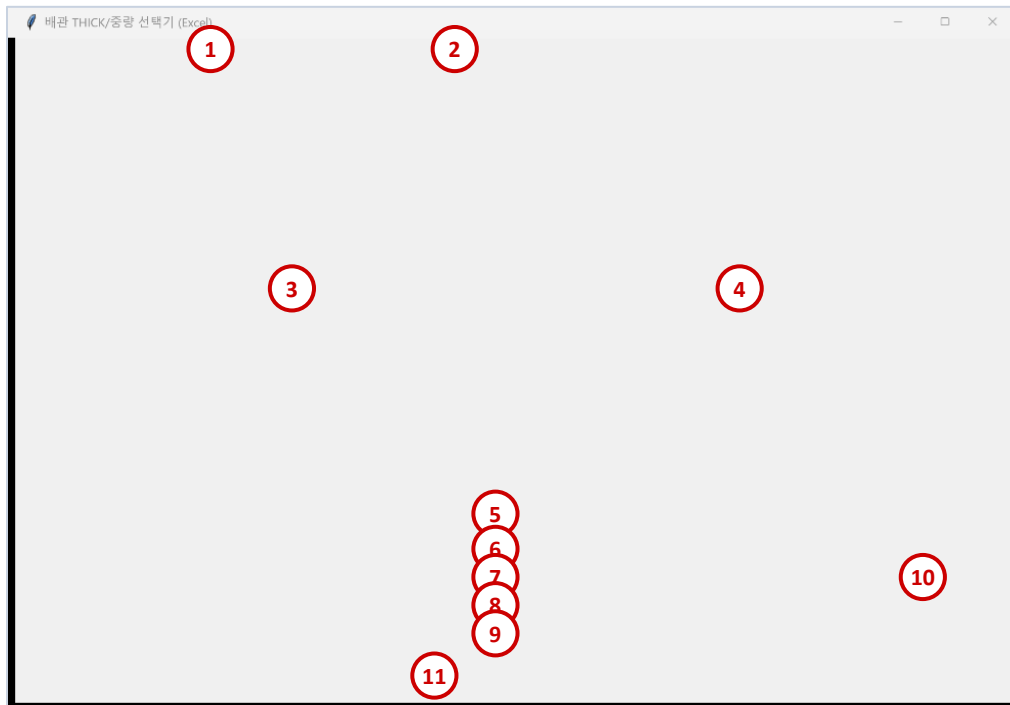
3. Two clicks give the answer

– Choose an item, click a nominal size and a schedule; thickness and weight fill in with a copy button

→ **Effect:** Values are read consistently from the source table and pasted straight into BOMs or notes, error-free

Required inputs : A folder of Excel pipe spec tables (*.xls/*.xlsx) with size, SCH, thickness and weight columns

Main Window – Item Guide



1. Item selector (dropdown of scanned Excel files)
2. Re-scan folder for Excel spec files
3. Nominal size (Size) button panel
4. Schedule (SCH) button panel
5. Result: Item
6. Result: Nominal size
7. Result: SCH
8. Result: Thickness (mm)
9. Result: Weight (kg/m or kg/EA)
10. Copy result value to clipboard
11. Note: weight column name kept as in source

Overview

What it does

1. Small field calculators: fitting weight
2. Pressure drop (steam/water/general)
3. Pipe thickness
4. Insulation density

Category: Calculation · Utility

Folder: ACEUTIL

Main scripts: (component scripts)

Scripts / Status: 6 scripts · Operational

Tags: 계산 · 압력강하 · 단열 · Tkinter

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Calculation · Utility.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- A set of component scripts — run the relevant step from the launcher / script list.
- Output samples and the program window above reflect the current build.

Thank You

Piping Calculators (ACEUTIL) · Plant Engineering Total Service System